# Intervention-relative control selectors in an open Bose–Hubbard chain: variance, occupation, and the transport boundary

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We ask whether a local observable can serve as a *targeted control handle* in an open quantum system, and whether the answer depends on the class of intervention applied. Using exact Lindblad evolution of a one-dimensional open Bose–Hubbard chain ($L \leq 8$), we show that the local occupation variance $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ identifies the sites whose targeted perturbation produces the largest finite-horizon response—but only for *number-conserving* transport interventions. The handle is active precisely when the selected sites enter a *directed self-draining* regime, which by particle-number continuity is the time-integrated net outward current; the burn-in current divergence of the selected set predicts the response ($\rho \approx 0.9$ at $L = 6, 7, 8$), and the coupling-driven crossover is a current reversal. The previously proposed redistribution-susceptibility explanation is shown to be non-discriminating (correlation $\approx 0$ with success) and is retired. Under deterministic tilt and random disorder $F_i$ outperforms geometry; it generalizes to coherent local detuning (a sign-controlled handle) but *not* to particle loss, where local occupation $\langle \hat{n}_i \rangle$ is the operative selector; and direct bond (hopping) control does not reveal a deeper bond-level handle. The result is a clean, falsifiable map: $F_i$ is the selector for $N$-conserving transport-control handles and $\langle \hat{n}_i \rangle$ for the $N$-changing loss handle, with the boundary set by particle-number conservation.

## I. INTRODUCTION

The interplay of tunnelling and on-site interaction in the Bose–Hubbard model underlies much of cold-atom many-body physics [1–4], and coupling such systems to an environment—most simply through local dephasing in the Lindblad framework [5–9]—drives them into spatially heterogeneous, far-from-equilibrium states. The deliberate use of dissipation as a control and state-preparation resource is by now a broad program [10–13]. A natural and underexplored question in this setting is *operational*: can a local structural observable act as a control handle—does perturbing the sites it selects produce a measurably different future than perturbing the same budget elsewhere—and is such "control relevance" a fixed property of the observable or a property of the observable *together with* the intervention?

This paper answers both questions for one concrete observable, the local occupation variance $F_i$. An earlier version of this study established only that targeting high-$F_i$ sites with extra dephasing beats matched-budget random targeting in a "positive pocket" of couplings. Here we (i) identify the dynamical mechanism, (ii) show it is not the redistribution susceptibility previously invoked, (iii) separate it from geometry under symmetry breaking, and (iv) map how the operative selector changes as the intervention class is varied—dephasing, coherent detuning, particle loss, and direct bond control. The central finding is that description privilege is *intervention-relative*: $F_i$ is the right selector for number-conserving transport handles, while local occupation takes over for number-changing loss. All results are exact (no Trotter, unravelling, or mean-field); the scope is correspondingly limited to small chains, $L \leq 8$.

## II. MODEL AND METHODS

The Hamiltonian for $L$ sites with open boundaries is

$$\hat{H} = -J \sum_i (\hat{a}_i^\dagger \hat{a}_{i+1} + \text{h.c.}) + \frac{U}{2} \sum_i \hat{n}_i(\hat{n}_i - 1) \left( + \sum_i \mu_i \hat{n}_i \right), \tag{1}$$

at half filling $N = \lfloor L/2 \rfloor$, local cutoff $n_{\max} = 3$; the optional $\mu_i$ implements tilt/disorder. The density matrix obeys $\dot{\hat{\rho}} = -i[\hat{H}, \hat{\rho}] + \sum_i \gamma_i (\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \frac{1}{2}\{\hat{L}_i^\dagger \hat{L}_i, \hat{\rho}\})$, evolved exactly via the action of the matrix exponential [14]. Baseline dephasing $\hat{L}_i = \hat{n}_i$ at $\gamma = 0.1$; the ground state is evolved for a burn-in $t_{\text{burn}} = 5$ to give $\hat{\rho}_{\text{burn}}$, on which the selector $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ is evaluated and the top-$k$ sites ($k = \lceil L/3 \rceil$) are selected.

*Interventions* (applied after burn-in) define four classes: extra dephasing ($\hat{L}_i = \hat{n}_i$, $N$-conserving, incoherent), coherent detuning ($\hat{H} \to \hat{H} + \mu_{\text{extra}} \sum_{i \in S} \hat{n}_i$, $N$-conserving, coherent), local loss ($\hat{L}_i = \hat{a}_i$, $N$-changing), and bond modulation ($J_b \to J_b(1 + \delta)$, $N$-conserving, coherent). Loss requires the multi-$N$ space $N \in \{0, \ldots, N_0\}$, which is exact (no truncation; $D = 84$ for $L = 6$) and reproduces the fixed-$N$ evolution to machine precision at $\gamma_{\text{loss}} = 0$.

*Target and sign convention.* For a selected set $S$ the directed self-drain is $D_S(\tau) = -\sum_{i \in S} \Delta \langle \hat{n}_i \rangle(\tau)$, so $D_S > 0$ means the selected sites lose occupation. The Hermitian bond current is $\hat{I}_{i,i+1} = -iJ(\hat{a}_i^\dagger \hat{a}_{i+1} - \text{h.c.}) = 2J \, \text{Im} \langle \hat{a}_i^\dagger \hat{a}_{i+1} \rangle$; because the dephasing dissipator conserves each $\langle \hat{n}_i \rangle$ exactly, continuity holds, $d\langle \hat{n}_i \rangle / dt = \langle \hat{I}_{i-1,i} \rangle - \langle \hat{I}_{i,i+1} \rangle$ (verified numerically to $< 10^{-4}$),

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TABLE I. Current predictor of the handle, clean chain. $\rho$ is Spearman over all $(J/U, \tau)$ conditions.

| $L$ | onset $J/U$ ($\tau$=3) | $\rho(C_S^{\mathrm{burn}}, D_S)$ | $\rho(C_S^{\mathrm{burn}}, \text{handle})$ | max \|cont. err\| |
|---|---|---|---|---|
| 6 | 0.24 | +0.91 | +0.94 | $< 10^{-4}$ |
| 7 | 0.30 | +0.91 | +0.95 | $< 10^{-4}$ |
| 8 | 0.30 | +0.89 | +0.92 | $< 10^{-4}$ |

TABLE II. The redistribution-susceptibility correlation does not predict handle success ($\rho \approx 0$), though its per-condition value stays high in both low-coupling and pocket regimes.

| $L$ | $\rho$(handle, $F_i \leftrightarrow \chi^{\mathrm{redist}}$) | $\langle$corr$\rangle$ low-$J/U$ | $\langle$corr$\rangle$ pocket |
|---|---|---|---|
| 6 | −0.03 | 0.85 | 0.81 |
| 7 | +0.09 | 0.65 | 0.80 |
| 8 | −0.26 | 0.71 | 0.76 |
| disorder | +0.02 | – | – |

so $D_S$ *is* the time-integrated net outward current of $S$. The burn-in current divergence of $S$ is $C_S^{\mathrm{burn}} = \sum_{i\in S}(\langle \hat{I}_{i,i+1}\rangle - \langle \hat{I}_{i-1,i}\rangle)$, evaluated on $\hat{\rho}_{\mathrm{burn}}$. The handle is quantified by the percentile of the selected subset among all $\binom{L}{k}$ size-$k$ subsets, ranked by clipped local loss (or, for loss, by total system loss $\Delta N_{\mathrm{tot}} = N_0 - \langle \hat{N} \rangle$). Symmetry between $F_i$ and geometry is broken by a deterministic linear tilt and by on-site disorder $\mu_i \sim \mathrm{Uniform}(-\mu_{\max}, \mu_{\max})$.

TABLE III. Geometry separation for both $N$-conserving channels. "low-overlap" restricts to disorder realizations with $F_i$–geometry overlap $< 0.5$ in the pocket.

| channel | breaking | overlap | handle pct $F_i$ vs geo (pocket) | low-overlap $F$ |
|---|---|---|---|---|
| dephasing | tilt | 0.29 | 96.5 vs 63.7 | – |
| dephasing | disorder | 0.52 | 95.4 vs 72.1 | 95.6 vs 3 |
| detuning | tilt | – | 68.5 vs 44.6 | – |
| detuning | disorder | – | 70.9 vs 58.8 | 49.8 vs 3 |

## III. RESULTS

### A. The handle and its regimes

Targeting the highest-$F_i$ sites becomes effective above a coupling-dependent onset: at $\tau = 3$ the top-$F_i$ subset rises from below the median at $J/U = 0.12$ to the top of the all-subset distribution by $J/U \simeq 0.24$–$0.30$ for $L = 6, 7, 8$ [Fig. 1(a)]. The exhaustive percentiles reproduce those reported previously (e.g. $L = 8$, $J/U = 0.20$ gives 34/68/71 at $\tau = 1, 2, 3$), establishing that the same effect is under study. The remainder of the paper asks what variable controls this onset.

### B. Mechanism: directed self-drain is outward current

By continuity $D_S$ equals the integrated net outward current of $S$, and the static burn-in current divergence $C_S^{\mathrm{burn}}$ predicts both $D_S$ and the handle. Table I shows $\rho(C_S^{\mathrm{burn}}, D_S) \approx 0.9$ and $\rho(C_S^{\mathrm{burn}}, \text{handle}) \approx 0.94$ at all three sizes; Fig. 1(b) shows $C_S^{\mathrm{burn}}$ reversing sign (inward/fill $\to$ outward/drain) at the onset, and Fig. 2 the $C_S^{\mathrm{burn}}$–$D_S$ correlation. The mechanism is therefore transport, not a static local property: high-$F_i$ sites become useful once they sit on an outward-current channel, and that channel turns on with coupling.

### C. The previous explanation, retired

The earlier account attributed the handle to $F_i$ tracking a redistribution susceptibility. That correlation is real but *non-discriminating*: it is comparably large in every regime, including where the handle fails, so its correlation with actual success is $\approx 0$ (Table II). We therefore retire it as the mechanism (retaining it only as a necessary-but-not-sufficient fact). A naive "transport freezes at low $J/U$" picture is likewise not supported: the static bond coherence varies only weakly across the studied range, so the operative change is the *direction* of the response, captured by $C_S^{\mathrm{burn}}$, not the presence of transport.

### D. Beyond geometry: dephasing under symmetry breaking

In the clean reflection-symmetric chain $F_i$ coincides with geometric centrality, so the two must be separated by breaking the symmetry. Under both deterministic tilt and random disorder the top-$F_i$ selector beats a geometric selector (Table III, Fig. 3); the separation is sharpest where $F_i$ and geometry pick the most different sites (disorder, within-shell overlap $< 0.5$: 95.6 vs 37.3). Directed self-drain predicts the handle within the disorder ensemble ($\rho \simeq 0.65$ over all rows). The crossover location is not universal: strong tilt or disorder supplies directed current at all couplings, so the control parameter is the directed current itself, not $J/U$.

### E. Generality: coherent detuning

Replacing the dissipative dephasing handle with a coherent local detuning $\hat{H} \to \hat{H} + \mu_{\mathrm{extra}} \sum_{i\in S} \hat{n}_i$ yields a sign-controlled handle [$+\mu$ drains, $-\mu$ fills; Fig. 4(a)] that again beats geometry under symmetry breaking (Table III). Thus $F_i$ is a *general* $N$-conserving transport-control selector, not a dephasing-specific one. The mechanistic predictor, however, bifurcates: for dephasing $C_S^{\mathrm{burn}}$ *reads* the pre-existing current ($\rho \approx 0.7$–$0.9$), whereas detuning *imposes* its own current and $C_S^{\mathrm{burn}}$ is

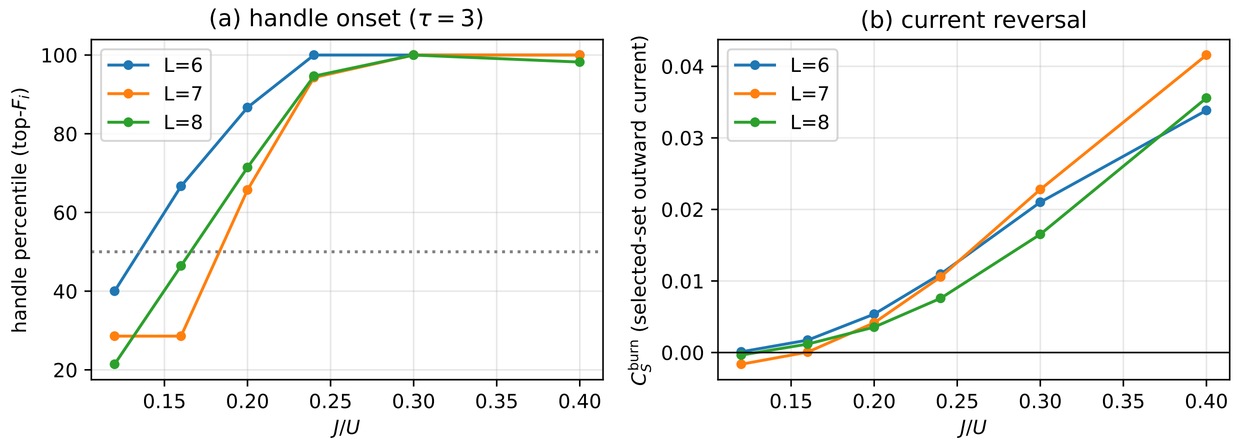


FIG. 1. **Directed self-drain and the current-reversal crossover.** (a) Handle percentile of the top-$F_i$ subset vs $J/U$ ($\tau = 3$, $L = 6, 7, 8$). (b) Burn-in outward current $C_S^{\rm burn}$ of the selected set vs $J/U$: it crosses zero (inward $\to$ outward) near the handle onset—the crossover is a current reversal.

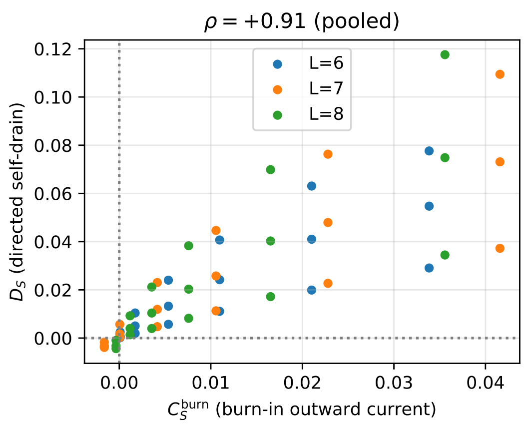


FIG. 2. Directed self-drain $D_S$ versus burn-in outward current $C_S^{\rm burn}$ (pooled $L = 6, 7, 8$); the two track each other, and both change sign together.

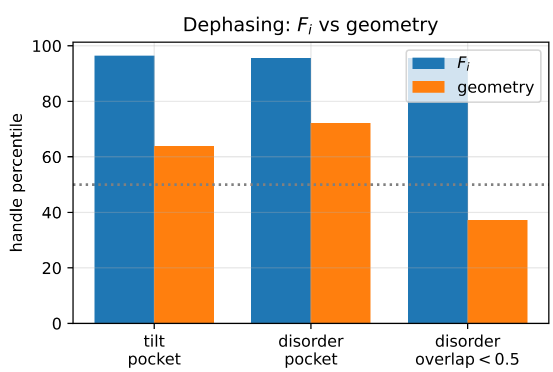


FIG. 3. Dephasing handle percentile, $F_i$ vs geometry, in the pocket and (disorder) where the two selectors most disagree. $F_i$ wins; geometry collapses below the median at low overlap.

TABLE IV. Loss is occupation-driven. Spearman of $\Delta N_{\rm tot}$ with each candidate selector; "separated" restricts to realizations where $F_i \neq$ maxn.

| predictor of $\Delta N_{\rm tot}$ | $\langle n_i \rangle$ | $F_i$ | current | geometry |
|---|---|---|---|---|
| clean $L = 6$ | +0.95 | +0.87 | +0.87 | +0.86 |
| sym-broken (separated) | +0.998 | +0.73 | – | +0.19 |

correspondingly weaker ($\rho \simeq 0.46$ tilt, 0.55 disorder).

### F. The boundary: particle loss

Local amplitude damping ($\hat{L}_i = \hat{a}_i$) changes $N$; the natural target is the total system loss $\Delta N_{\rm tot}$. Here $F_i$ is *not* operative: local occupation is the best predictor (Table IV, Fig. 5), decisively so under symmetry breaking, where $\rho(\Delta N_{\rm tot}, \langle \hat{n}_i \rangle) = +0.998$ versus $+0.73$ for $F_i$ in the realizations where the two selectors separate, and the maxn selector reaches the 99.6th percentile versus 84.9 for $F_i$. Physically, loss directly removes particles from occupied sites, so the operative variable is occupation; $F_i$'s residual loss-predictive power is only its clean-chain correlation with $\langle \hat{n}_i \rangle$. The boundary between the two selectors is thus *transport-modulation vs. particle-removal*—equivalently, particle-number conservation.

### G. Site versus bond control

Since current is a bond quantity, we tested whether the handle is fundamentally bond-level by modulating the hopping on selected bonds. Using the baseline-clean (induced) per-endpoint drain as the target, no bond observable predicts the effect (best $|\rho| = 0.19$); on the baseline-confounded total, the site-derived endpoint-$F_i$

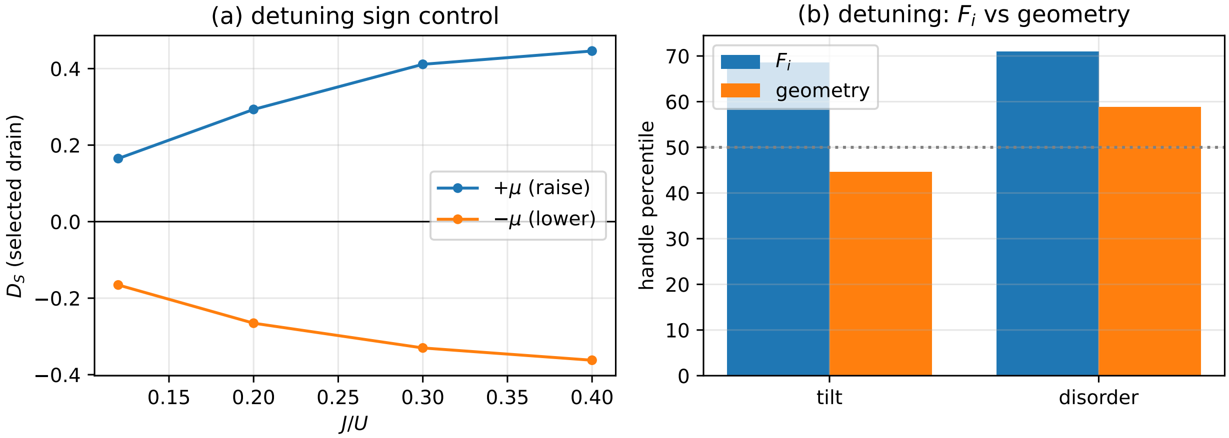


FIG. 4. **Coherent detuning.** (a) Selected-site drain $D_S$ vs $J/U$ for $\pm\mu$: raising the on-site energy drains, lowering fills—a sign knob absent for dephasing. (b) Detuning handle percentile, $F_i$ vs geometry, under tilt and disorder: $F_i$ wins.

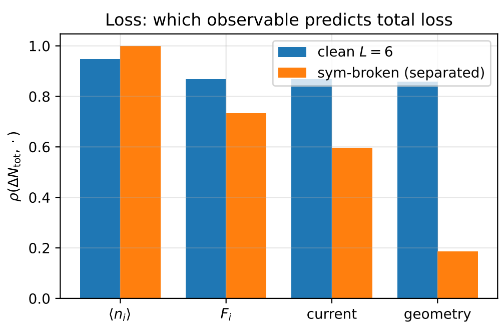


FIG. 5. Which observable predicts total particle loss. Occupation dominates; under symmetry breaking it is near-perfect while geometry is irrelevant.

TABLE V. Bond/hopping modulation. "induced" is the per-endpoint drain change due to the modulation (the clean bond-control signal); "total" is baseline-confounded.

| bond selector | $\rho$ (induced) | $\rho$ (total) |
|---|---|---|
| endpoint-$F_i$ (site-derived) | $-0.10$ | $+0.87$ |
| bond coherence | $+0.00$ | $+0.60$ |
| $\vert$bond current$\vert$ | $+0.07$ | $-0.83$ |
| endpoint occupation | $-0.19$ | $+0.86$ |

is not beaten by bond coherence or bond current (Table V). The site-level description is therefore kept: bond control does not expose a deeper bond-level handle, and the strong $F_i$–bond-coherence correlation is a correlation, not a better selector.

### H. Observer-geometry reanalysis (supporting)

A separate pre-registered reanalysis of the same simulation data, using an observer-geometry framework, provides supporting structure. Two outcomes are earned and reported: observer subspaces converge monotonically as the chain thermalizes, and the choice of scoring rule changes which site is "optimal" in every cell—both consistent with the redistribution picture. A third, earlier "predictive–causal dissociation" result is *not* retained as a finding: its headline rate ($\sim 0.90$) is dominated by a self-correlation artifact (the prediction target lies inside the observer family), and the honest rate once that observer is excluded is $\sim 0.52$. We report it here only as a limitation.

## IV. DISCUSSION

The results assemble into a single map (Table VI): the operative control selector is not a fixed property of the chain but depends on the allowed intervention. For number-conserving handles—incoherent dephasing and coherent detuning—the variance $F_i$ selects the transport-active, directed-self-draining sites, and the effect is mechanistically the integrated outward particle current. For the number-changing loss handle the operative selector is local occupation. Geometry is, in every case, at most a correlate. This is a concrete, falsifiable instance of *intervention-relative description privilege*: the same physical system selects different useful descriptions depending on the class of control applied, with the boundary set by a clean physical distinction (transport modulation versus particle removal). The picture connects naturally to dissipation engineering and to Zeno-type throttling of coherent transport [10, 13], while remaining agnostic about which microscopic relaxation mode carries the current—an explicit open question.

TABLE VI. The intervention-relative map: operative selector by intervention class.

| intervention | conserves $N$? | coherent? | **operative selector** |
|---|---|---|---|
| dephasing | yes | no | $F_i$ |
| detuning | yes | yes | $F_i$ |
| local loss | no | no | $\langle n_i \rangle$ |
| bond hopping mod. | yes | yes | **site-level $F_i$ (not bond)** |

## V. CONCLUSION

In exact small open Bose–Hubbard chains, local number variance is the operative control selector for number-conserving transport interventions (dephasing and detuning), while local occupation is the operative selector for number-changing loss; geometry is not fundamental; and the handle operates only in the directed self-draining regime, which is the integrated outward particle current, with the coupling crossover a current reversal. The natural next steps—a Liouvillian spectral decomposition to identify which slow mode carries the current, and approximate methods (quantum trajectories, matrix-product operators) to test persistence at larger $L$—are left for future work. We make no claim beyond the exact, small-$L$ regime studied here: not the thermodynamic limit, not a universal control law, and not laboratory realization.

## Appendix A: Numerical validation

Time evolution is exact via Krylov/Padé matrix-exponential action; the sparse and dense code paths agree to $\sim 10^{-15}$. For loss, the multi-$N$ construction is exact (it includes all sectors $N \le N_0$ reachable under loss), reproduces the fixed-$N$ occupations to $2 \times 10^{-16}$ at $\gamma_{\rm loss} = 0$, preserves trace and positivity, and gives a particle number that decreases monotonically and linearly in $\gamma_{\rm loss}$. Continuity $d\langle \hat{n}_i \rangle/dt = \langle \hat{I}_{i-1,i} \rangle - \langle \hat{I}_{i,i+1} \rangle$ is verified to $< 10^{-4}$ (finite difference), and the per-bond-$J$ Hamiltonian reduces to the scalar-$J$ Hamiltonian when all $J_b$ are equal. All headline numbers are computed directly from the deposited simulation outputs; code and data are available with the manuscript.


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